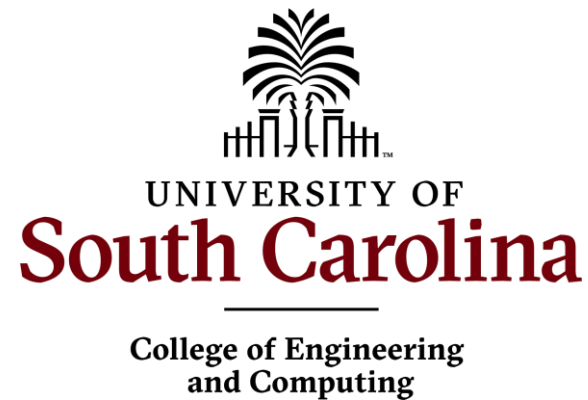


INTRODUCTION TO APPLIED NUMERICAL METHODS

Gauss Elimination

Lang Yuan, Associate Professor



REPRESENTING LINEAR ALGEBRAIC EQUATIONS IN MATRIX FORM

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n &= b_n \end{aligned}$$

Linear algebraic equations

$$\Rightarrow [A][X] = [B]$$

Matrix Form

$$[A] = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}$$

[A] is the n by n square matrix of coefficients

$$[X] = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix}$$

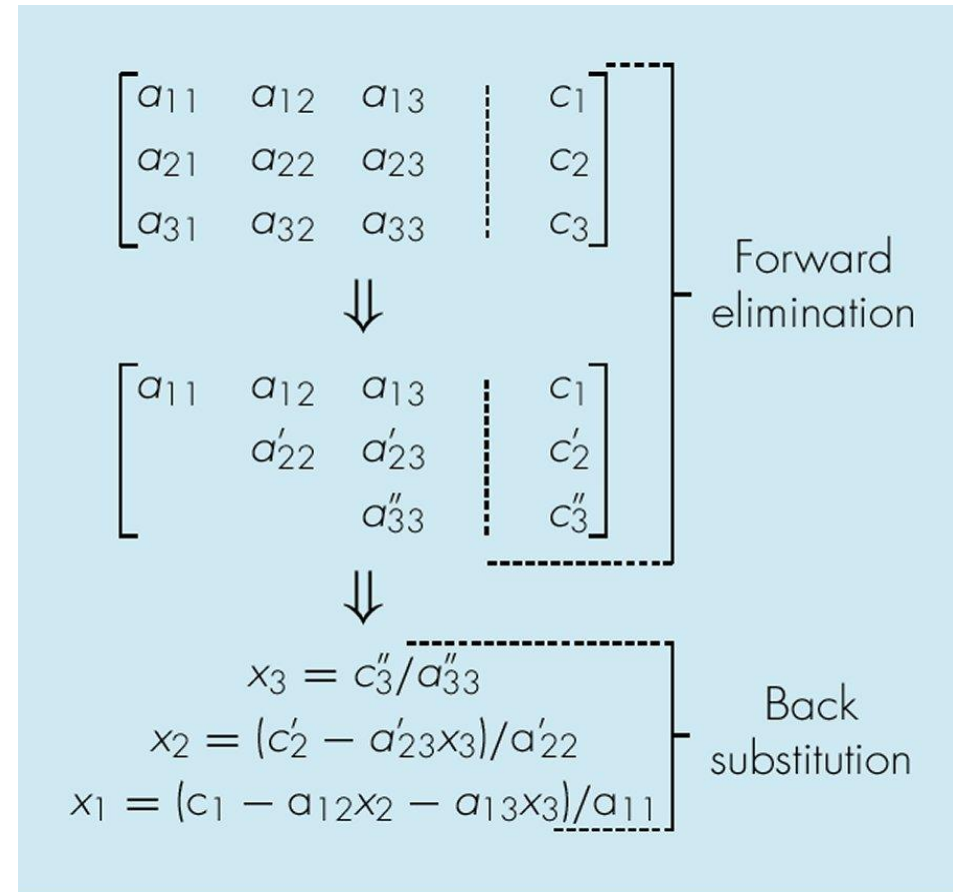
[X] is the n by 1 column vector of unknowns

$$[B] = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_{n-1} \\ b_n \end{bmatrix}$$

[B] is the n by 1 column vector of constants

NAIVE GAUSS ELIMINATION

- Extension of *method of elimination* to large sets of equations by developing a systematic scheme or algorithm to eliminate unknowns and to back substitute.
 - Forward elimination of unknowns
 - Back substitution



NAIVE GAUSS ELIMINATION

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} b_1 \\ b_2 \\ b_3 \end{Bmatrix} \quad \begin{matrix} (1) \\ (2) \\ (3) \end{matrix}$$

- General process for a 3 by 3 matrix
- Forward elimination
 - Keep the first equation and reconstruct the second and until the last one
 - Keep the second equation and reconstruct the third one until the last one
 -
 - Elimination :
 - $(2)^1 = (2) - (1) * \frac{a_{21}}{a_{11}} \rightarrow \left(a_{22} - \frac{a_{12}a_{21}}{a_{11}}\right)x_2 + \left(a_{23} - \frac{a_{13}a_{21}}{a_{11}}\right)x_3 = b_2 - \frac{b_1a_{21}}{a_{11}}$
 - $(3)^1 = (3) - (1) * \frac{a_{31}}{a_{11}} \rightarrow \left(a_{32} - \frac{a_{12}a_{31}}{a_{11}}\right)x_2 + \left(a_{33} - \frac{a_{13}a_{31}}{a_{11}}\right)x_3 = b_3 - \frac{b_1a_{31}}{a_{11}}$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & \left(a_{22} - \frac{a_{12}a_{21}}{a_{11}}\right) & \left(a_{23} - \frac{a_{13}a_{21}}{a_{11}}\right) \\ 0 & \left(a_{32} - \frac{a_{12}a_{31}}{a_{11}}\right) & \left(a_{33} - \frac{a_{13}a_{31}}{a_{11}}\right) \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} b_1 \\ b_2 - \frac{b_1a_{21}}{a_{11}} \\ b_3 - \frac{b_1a_{31}}{a_{11}} \end{Bmatrix}$$

NAIVE GAUSS ELIMINATION

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a_{22}^1 & a_{23}^1 \\ 0 & a_{32}^1 & a_{33}^1 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} b_1 \\ b_2^1 \\ b_3^1 \end{Bmatrix} \quad \begin{matrix} (1) \\ (2)^1 \\ (3)^1 \end{matrix}$$

$$a_{22}^1 = \left(a_{22} - \frac{a_{12}a_{21}}{a_{11}} \right)$$

$$a_{23}^1 = \left(a_{23} - \frac{a_{13}a_{21}}{a_{11}} \right)$$

$$a_{32}^1 = \left(a_{32} - \frac{a_{12}a_{31}}{a_{11}} \right)$$

$$a_{33}^1 = \left(a_{33} - \frac{a_{13}a_{31}}{a_{11}} \right)$$

$$b_2^1 = b_2 - \frac{b_1a_{21}}{a_{11}}$$

$$b_3^1 = b_3 - \frac{b_1a_{31}}{a_{11}}$$

- Forward elimination

- $$(3)^2 = (3)^1 - (2)^1 * \frac{a_{32}^1}{a_{22}^1} \rightarrow \left(a_{33}^1 - \frac{a_{23}^1a_{32}^1}{a_{22}^1} \right) x_3 = b_3^1 - \frac{b_2^1a_{32}^1}{a_{22}^1}$$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a_{22}^1 & a_{23}^1 \\ 0 & 0 & a_{33}^2 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} b_1 \\ b_2^1 \\ b_3^2 \end{Bmatrix}$$

$$a_{33}^2 = \left(a_{33}^1 - \frac{a_{23}^1a_{32}^1}{a_{22}^1} \right)$$

$$b_3^2 = b_3^1 - \frac{b_2^1a_{32}^1}{a_{22}^1}$$

- Get the solution: Back substitution

NAIVE GAUSS ELIMINATION

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} b_1 \\ b_2 \\ b_3 \end{Bmatrix}$$

$$[A]\{x\} = \{b\}$$



$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a_{22}^1 & a_{23}^1 \\ 0 & 0 & a_{33}^2 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} b_1 \\ b_2^1 \\ b_3^2 \end{Bmatrix}$$

$$[U]\{x\} = \{D\}$$

GAUSS ELIMINATION: AN EXAMPLE

Use Gauss elimination to solve

$$3x_1 - 0.1x_2 - 0.2x_3 = 7.85 \quad (1)$$

$$0.1x_1 + 7x_2 - 0.3x_3 = -19.3 \quad (2)$$

$$0.3x_1 - 0.2x_2 + 10x_3 = 71.4 \quad (3)$$

Solution: Forward Elimination

Step 1.1:

Eq. (1) $\times (0.1/3)$ ① $\times \left(\frac{0.1}{3} \right)$

$$0.1x_1 - \frac{0.01}{3}x_2 - \frac{0.02}{3}x_3 = 0.2617 \quad (1^*)$$

Subtract Eq. (1*) from Eq. (2)

$$0.1x_1 + 7x_2 - 0.3x_3 = -19.3 \quad (2)$$

$$0.1x_1 - \frac{0.01}{3}x_2 - \frac{0.02}{3}x_3 = 0.2617 \quad (1^*)$$

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$$7.00333x_2 - 0.293333x_3 = -19.5617 \quad (2')$$

New Equation Set

$$3x_1 - 0.1x_2 - 0.2x_3 = 7.85 \quad (1)$$

$$7.00333x_2 - 0.293333x_3 = -19.5617 \quad (2')$$

$$0.3x_1 - 0.2x_2 + 10x_3 = 71.4 \quad (3)$$

GAUSS ELIMINATION: AN EXAMPLE

Solution: Forward Elimination

Step 1.2:

$$\text{Eq. (1)} \times (0.3/3) \quad \textcircled{1} \times \left(\frac{0.3}{3}\right)$$

$$0.3x_1 - 0.01x_2 - 0.02x_3 = 0.785 \quad (1^+)$$

Subtract Eq. (1⁺) from Eq. (3)

$$0.3x_1 - 0.2x_2 + 10x_3 = 71.4 \quad (3)$$

—

$$0.3x_1 - 0.01x_2 - 0.02x_3 = 0.785 \quad (1^+)$$



$$-0.190000x_2 + 10.0200x_3 = 70.6150 \quad (3')$$



New Equation Set

$$3x_1 - 0.1x_2 - 0.2x_3 = 7.85 \quad (1)$$

$$7.00333x_2 - 0.293333x_3 = -19.5617 \quad (2')$$

$$0.3x_1 - 0.2x_2 + 10x_3 = 71.4 \quad (3)$$

New Equation Set

$$3x_1 - 0.1x_2 - 0.2x_3 = 7.85 \quad (1)$$

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$$-0.190000x_2 + 10.0200x_3 = 70.6150 \quad (3')$$

GAUSS ELIMINATION: AN EXAMPLE

New Equation Set

$$\begin{array}{rcl} 3x_1 & -0.1x_2 & -0.2x_3 = 7.85 & (1) \\ & 7.00333x_2 - 0.293333x_3 = -19.5617 & (2') \\ & -0.190000x_2 + 10.0200x_3 = 70.6150 & (3') \end{array}$$

Solution: Forward Elimination

Step 2:

Eq. (2') $\times (-0.19/7.00333)$ $\textcircled{2}$ $\times \left(\frac{-0.19}{7.00333} \right)$

$$-0.19x_2 + 0.0079581x_3 = 0.5307 \quad (2^*)$$

Subtract Eq. (2^{*}) from Eq. (3')

$$-0.190000x_2 + 10.0200x_3 = 70.6150 \quad (3')$$

—

$$-0.19x_2 + 0.0079581x_3 = 0.5307 \quad (2^*)$$



$$10.0120x_3 = 70.0843 \quad (3'')$$



New Equation Set

$$\begin{array}{rcl} 3x_1 & -0.1x_2 & -0.2x_3 = 7.85 \\ & 7.00333x_2 - 0.293333x_3 = -19.5617 \\ & & 10.0120x_3 = 70.0843 \end{array}$$

GAUSS ELIMINATION: AN EXAMPLE

New Equation Set

$$3x_1 - 0.1x_2 - 0.2x_3 = 7.85 \quad (1)$$

$$7.00333x_2 - 0.293333x_3 = -19.5617 \quad (2')$$

$$10.0120x_3 = 70.0843 \quad (3'')$$

Solution: **Back Substitution**

Eq. (3'') can be solved for

$$x_3 = \frac{70.0843}{10.0120} = 7.0000$$

Back-substitute x_3 to Eq. (2')

$$7.00333x_2 - 0.293333(7.0000) = -19.5617$$

$$x_2 = \frac{-19.5617 + 0.293333(7.0000)}{7.00333} = -2.50000$$

GAUSS ELIMINATION: AN EXAMPLE

$$\begin{array}{rcl} 3x_1 & -0.1x_2 & -0.2x_3 = 7.85 & (1) \\ & 7.00333x_2 - 0.293333x_3 & = -19.5617 & (2') \\ & & 10.0120x_3 = 70.0843 & (3'') \end{array}$$

Solution: **Back Substitution**

Back-substitute x_2 and x_3 to Eq. (1)

$$3x_1 - 0.1x_2 - 0.2x_3 = 7.85$$



$$3x_1 - 0.1(-2.50000) - 0.2(7.0000) = 7.85$$



$$x_1 = \frac{7.85 + 0.1(-2.50000) + 0.2(7.0000)}{3} = 3.00000$$

GAUSS ELIMINATION: AN EXAMPLE

Use Gauss elimination to solve

$$3x_1 - 0.1x_2 - 0.2x_3 = 7.85 \quad (1)$$

$$0.1x_1 + 7x_2 - 0.3x_3 = -19.3 \quad (2)$$

$$0.3x_1 - 0.2x_2 + 10x_3 = 71.4 \quad (3)$$

With six significant figures during the computation

Verification:

$$x_1 = 3, x_2 = -2.5, \text{ and } x_3 = 7$$



$$3(3) - 0.1(-2.5) - 0.2(7) = 7.85$$

$$0.1(3) + 7(-2.5) - 0.3(7) = -19.3$$

$$0.3(3) - 0.2(-2.5) + 10(7) = 71.4$$

Basic Steps:

- Elimination of unknowns
- Solution through back substitution

$$\begin{aligned}0.0003x_1 + 3.0000x_2 &= 2.0001 \\1.0000x_1 + 1.0000x_2 &= 1.0000\end{aligned}$$

PITFALLS OF ELIMINATION METHODS

- *Division by zero.* It is possible that during both elimination and back-substitution phases a division by zero can occur.

- Elimination: (Eq 1.) $\times \left(\frac{a_{21}}{a_{11}} \right)$
- Pivoting: rearrange the largest element first

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 &= b_1 & \text{Eq.1} \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 &= b_2 & \text{Eq.2} \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 &= b_3 & \text{Eq.3} \end{aligned}$$

- *Round-off errors*

$$\begin{aligned} 0.0003x_1 + 3.0000x_2 &= 2.0001 \\ 1.0000x_1 + 1.0000x_2 &= 1.0000 \end{aligned}$$

- $x_2 = \frac{2}{3}$
- $x_1 = (2.0001 - 3 * x_2)/0.0003$

x2	x1	Error
0.667	-3.33	1099
0.6667	0	100
0.66667	0.3	10

PITFALLS OF ELIMINATION METHODS

- *Ill-conditioned systems.* Systems where small changes in coefficients result in large changes in the solution. Alternatively, it happens when two or more equations are nearly identical, resulting a wide ranges of answers to approximately satisfy the equations. Since round off errors can induce small changes in the coefficients, these changes can lead to large solution errors.

